

► RENAL—PATHOLOGY

Casts in urine

Presence of casts indicates that hematuria/pyuria is of glomerular or renal tubular origin.
 Bladder cancer, kidney stones → hematuria, no casts.
 Acute cystitis → pyuria, no casts.
 All casts contain a matrix composed primarily of Tamm-Horsfall mucoprotein (uromodulin), secreted by renal tubular cells to prevent UTIs.

RBC casts **A**

Glomerulonephritis, hypertensive emergency.

WBC casts

Tubulointerstitial inflammation, acute pyelonephritis, transplant rejection.

Granular casts **B**

Acute tubular necrosis (ATN). Can be “muddy brown” in appearance.

Fatty casts (“oval fat bodies”)

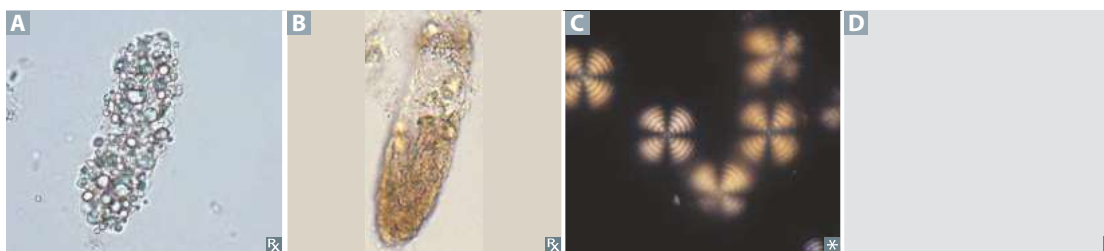
Nephrotic syndrome. Associated with “Maltese cross” sign **C**.

Waxy casts

End-stage renal disease/chronic kidney disease.

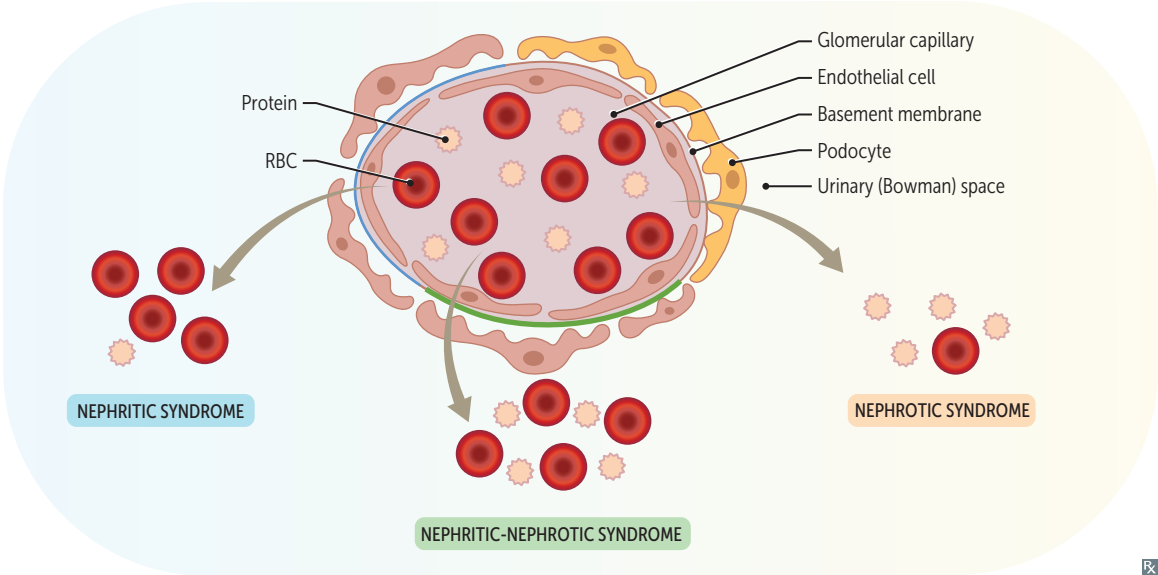
Hyaline casts **D**

Nonspecific, can be a normal finding with dehydration, exercise, or diuretic therapy.

**Nomenclature of glomerular disorders**

TYPE	CHARACTERISTICS	EXAMPLE
Focal	< 50% of glomeruli are involved	Focal segmental glomerulosclerosis
Diffuse	> 50% of glomeruli are involved	Diffuse proliferative glomerulonephritis
Proliferative	Hypercellular glomeruli	Membranoproliferative glomerulonephritis
Membranous	Thickening of GBM	Membranous nephropathy
Primary glomerular disease	1° disease of the kidney specifically impacting the glomeruli	Minimal change disease
Secondary glomerular disease	Systemic disease or disease of another organ system that also impacts the glomeruli	SLE, diabetic nephropathy

Glomerular diseases



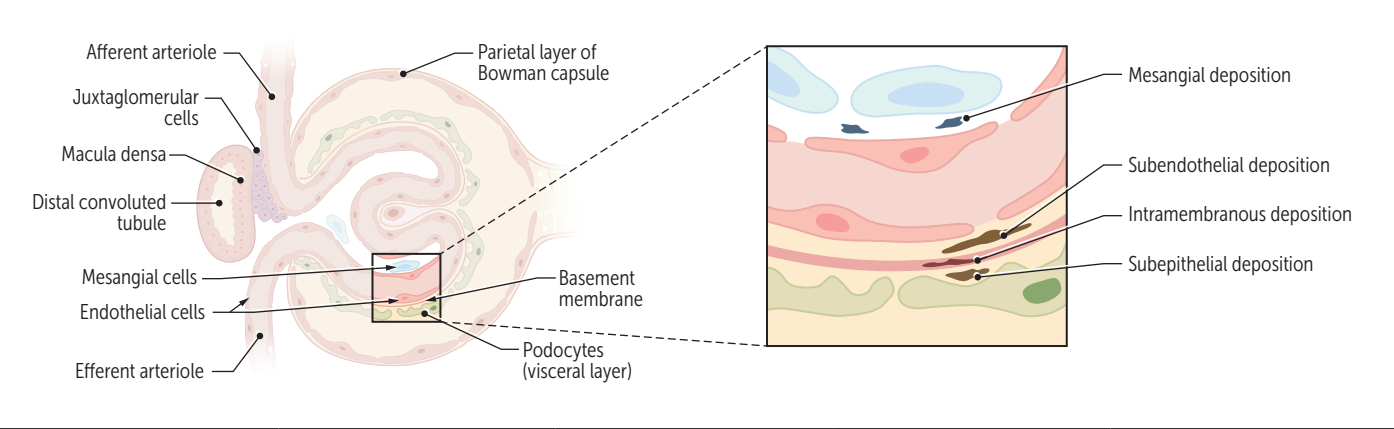
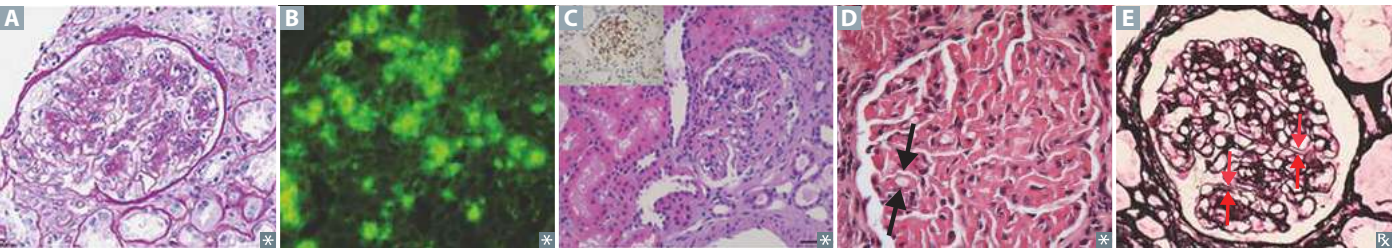
TYPE	ETIOLOGY	CLINICAL PRESENTATION	EXAMPLES
Nephritic syndrome	Glomerular inflammation → GBM damage → loss of RBCs into urine → dysmorphic RBCs, hematuria	Hematuria, RBC casts in urine ↓ GFR → oliguria, azotemia ↑ renin release, HTN Proteinuria often in the subnephrotic range (< 3.5 g/day) but in severe cases may be in nephrotic range	▪ Infection-associated glomerulonephritis ▪ Goodpasture syndrome ▪ IgA nephropathy (Berger disease) ▪ Alport syndrome ▪ Membranoproliferative glomerulonephritis
Nephrotic syndrome	Podocyte damage → impaired charge barrier → proteinuria	Massive proteinuria (> 3.5 g/day) with edema, hypoalbuminemia → ↑ hepatic lipogenesis → hypercholesterolemia Frothy urine with fatty casts Associated with hypercoagulable state (eg, renal vein thrombosis) due to antithrombin loss in urine and ↑ risk of infection (loss of IgGs in urine and soft tissue compromise by edema)	May be 1° (eg, direct podocyte damage) or 2° (podocyte damage from systemic process): ▪ Focal segmental glomerulosclerosis (1° or 2°) ▪ Minimal change disease (1° or 2°) ▪ Membranous nephropathy (1° or 2°) ▪ Amyloidosis (2°) ▪ Diabetic glomerulonephropathy (2°)
Nephritic-nephrotic syndrome	Severe GBM damage → loss of RBCs into urine + impaired charge barrier → hematuria + proteinuria	Nephrotic-range proteinuria (> 3.5 g/day) and concomitant features of nephritic syndrome	Can occur with any form of nephritic syndrome, but is most common with: ▪ Diffuse proliferative glomerulonephritis ▪ Membranoproliferative glomerulonephritis

Nephritic syndrome

	MECHANISM	LIGHT MICROSCOPY	IMMUNOFLUORESCENCE	ELECTRON MICROSCOPY
Infection-related glomerulonephritis	Type III hypersensitivity reaction with consumptive hypocomplementemia Children: seen ~2–4 weeks after group A streptococcal pharyngitis or skin infection Adults: <i>Staphylococcus</i> is additional causative agent	Enlarged and hypercellular glomeruli A	Granular (“starry sky”) appearance (“lumpy-bumpy”) B due to IgG, IgM, and C3 deposition along GBM and mesangium	Subepithelial IC humps
IgA nephropathy (Berger disease)	Occurs concurrently with respiratory or GI tract infections (IgA is secreted by mucosal linings) Renal pathology of IgA vasculitis	Mesangial proliferation	IgA-based IC deposits in mesangium	Mesangial IC deposition
Rapidly progressive (crescentic) glomerulonephritis	Poor prognosis Multiple causes: Type II HSR in Goodpasture syndrome	Crescent moon shape C ; crescents consist of fibrin and plasma proteins (eg, C3b) with glomerular parietal cells, monocytes, macrophages	Linear IF due to antibodies to GBM and alveolar basement membrane: Goodpasture syndrome—hematuria/hemoptysis Negative IF/Pauci-immune (no IgC3 deposition): granulomatosis with polyangiitis—PR3-ANCA/c-ANCA, eosinophilic granulomatosis with polyangiitis, or M icroscopic p olyangiitis— M PO-ANCA/ p -ANCA Granular IF—PSGN or DPGN	Goodpasture syndrome: breaks in GBM, necrosis and crescent formation with no deposits Pauci-immune: usually no deposits EM features depend on underlying cause
Diffuse proliferative glomerulonephritis	Often due to SLE (think “wire lupus”); DPGN and MPGN often present as nephritic and nephrotic syndromes concurrently	“Wire looping” of capillaries D	Granular	Subendothelial, sometimes subepithelial or intramembranous IgG-based ICs often with C3 deposition

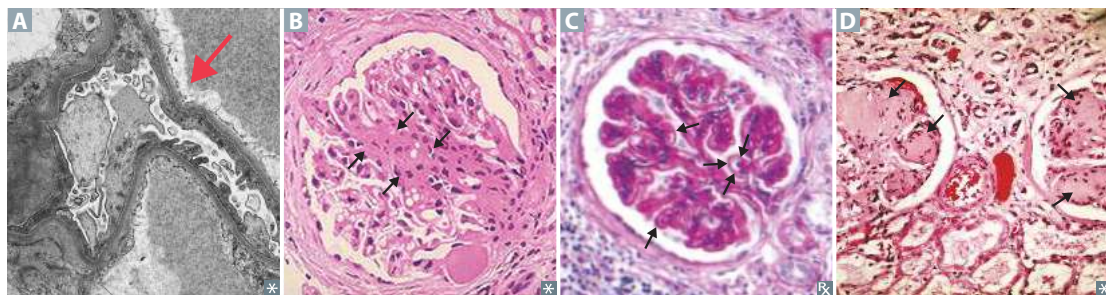
Nephritic syndrome (continued)

Alport syndrome	Type IV collagen mutation → GBM alterations; mostly X-linked dominant. Eye problems (eg, retinopathy, anterior lenticonus), glomerulonephritis, sensorineural hearing loss (can't see, can't pee, can't hear a bee)	Irregular thinning and thickening and splitting of GBM	Initially negative; nonspecific staining (usually stays negative)	"Basket-weave" appearance due to irregular thickening and longitudinal splitting of GBM
Membrano-proliferative glomerulonephritis	Type I may be 2° to HBV or HCV infection; type II associated with C3 nephritic factor (IgG autoantibody that stabilizes C3 convertase → persistent complement activation → ↓ C3)	Mesangial ingrowth → GBM splitting → "tram-track" on H&E and PAS E stains	Granular	Type I—subendothelial IC deposits Type II—intramembranous deposits, also called dense deposit disease



Nephrotic syndrome

	MECHANISM	LIGHT MICROSCOPY	IMMUNOFLUORESCENCE	ELECTRON MICROSCOPY
Minimal change disease	Often 1° (idiopathic), triggered by recent infection, immunization, immune stimulus (4 Is); rarely 2° to lymphoma (eg, cytokine-mediated damage). More common in children.	Normal glomeruli (lipid may be seen in PT cells)	⊖	Effacement of podocyte foot processes A
Focal segmental glomerulosclerosis	Can be 1° (idiopathic) or 2° (eg, HIV infection, sickle cell disease, obesity, or congenital malformations); may progress to CKD. More common in Black people.	Segmental sclerosis and hyalinosis B	Often ⊖ but may be ⊕ for nonspecific focal deposits of IgM, C3, C1	Effacement of podocyte foot processes
Membranous nephropathy	Also called membranous glomerulonephritis. Can be 1° (eg, antibodies to phospholipase A ₂ receptor) or 2° to drugs (eg, NSAIDs, penicillamine, gold), infections (eg, HBV, HCV, syphilis), SLE, or solid tumors. ↑ risk of thromboembolism (eg, DVT, renal vein thrombosis).	Diffuse capillary and GBM thickening C	Granular due to immune complex (IC) deposition	“Spike and dome” appearance of subepithelial deposits
Amyloidosis	Kidney most commonly involved organ. Associated with chronic conditions that predispose to amyloid deposition (eg, AL amyloid, AA amyloid, prolonged dialysis).	Congo red stain shows apple-green birefringence (polarized light) due to amyloid deposition in the mesangium	AL amyloidosis: may be positive for lambda and kappa light chains AA amyloidosis: positive for AA protein	Mesangial expansion by amyloid fibrils
Diabetic glomerulonephropathy	Most common cause of ESRD in United States. Hyperglycemia → nonenzymatic glycation of tissue proteins → mesangial expansion → GBM thickening and ↑ permeability. Hyperfiltration (glomerular HTN and ↑ GFR) → glomerular hypertrophy and glomerular scarring (glomerulosclerosis) → further progression of nephropathy. Look for albuminuria with ↑ urine albumin-to-creatinine ratio. ACE inhibitors and ARBs are renoprotective.	Mesangial expansion, GBM thickening, eosinophilic nodular glomerulosclerosis (Kimmelstiel-Wilson lesions D)	Non-specific staining. Usually negative.	Prominent thickening of GBM with expanded mesangium, predominantly due to increased mesangial matrix, segmental podocyte effacement



Kidney stones

Can lead to severe complications such as hydronephrosis, pyelonephritis, and acute kidney injury. Obstructed stone presents with unilateral flank tenderness, colicky pain radiating to groin, hematuria. Treat and prevent by encouraging fluid intake. Radiolucent stones: I can't **c** (see) **u** (you) (**c**ystine and **u**ric acid).

CONTENT	PRECIPITATES WITH	X-RAY FINDINGS	CT FINDINGS	URINE CRYSTAL	NOTES
Calcium	Calcium oxalate: hypocitraturia	Radiopaque	Hyperdense	Shaped like envelope A or dumbbell	Calcium stones most common (80%); calcium oxalate more common than calcium phosphate stones. Can result from ethylene glycol (antifreeze) ingestion, vitamin C overuse, hypocitraturia (usually associated with ↓ urine pH), malabsorption (eg, Crohn disease). Treatment: thiazides, citrate, low-sodium diet.
	Calcium phosphate: ↑ pH	Radiopaque	Hyperdense	Wedge-shaped prism	Treatment: low-sodium diet, thiazides.
Ammonium magnesium phosphate (struvite)	↑ pH	Radiopaque	Hyperdense	Coffin lid ("sarcophagus")	Account for 15% of stones. Caused by infection with urease ⊕ bugs (eg, <i>Proteus mirabilis</i> , <i>Staphylococcus saprophyticus</i> , <i>Klebsiella</i>) that hydrolyze urea to ammonia → urine alkalinization. Commonly form staghorn calculi B . Treatment: eradication of underlying infection, surgical removal of stone.
Uric acid	↓ pH	Radiolucent	Visible	Rhomboid C or rosettes	About 5% of all stones. Risk factors: arid climates, acidic pH. Strong association with hyperuricemia (eg, gout). Often seen in diseases with ↑ cell turnover (eg, leukemia). Treatment: alkalinization of urine, allopurinol.
Cystine	↓ pH	Faintly radiopaque	Moderately radiodense	Hexagonal D	Hereditary (autosomal recessive) condition in which C ystine-reabsorbing PCT transporter loses function, causing cystinuria. Transporter defect also results in poor reabsorption of O rnithine, L ysine, A rginine (COLA). Cystine is poorly soluble, thus stones form in urine. Usually begins in childhood. Can form staghorn calculi. Sodium cyanide nitroprusside test ⊕. " S ixtine" stones have s ix sides. Treatment: low sodium diet, alkalinization of urine, chelating agents (eg, tiopronin, penicillamine) if refractory.

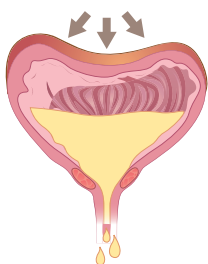
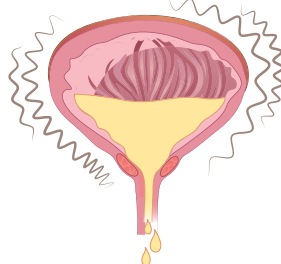
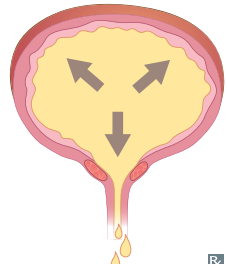


Hydronephrosis

Distention/dilation of renal pelvis and/or calyces **A**. Mostly caused by urinary tract obstruction (eg, urinary tract stones, severe BPH, congenital obstructions, locally advanced cervical cancer, injury to ureter); other causes include retroperitoneal fibrosis, vesicoureteral reflux. Dilation occurs proximal to site of pathology. Serum creatinine becomes elevated if obstruction is bilateral or if patient has an obstructed solitary kidney. Leads to compression and possible atrophy of renal cortex and medulla.

Urinary incontinence

Mixed incontinence has features of both stress and urgency incontinence.

	Stress incontinence	Urgency incontinence	Overflow incontinence
			
MECHANISM	Outlet incompetence (urethral hypermobility or intrinsic sphincter deficiency) → leak with ↑ intra-abdominal pressure (eg, sneezing, lifting) ⊕ bladder stress test (directly observed leakage from urethra upon coughing or Valsalva maneuver)	Detrusor overactivity → leak with urge to void immediately	Incomplete emptying (detrusor underactivity or outlet obstruction) → leak with overfilling, ↑ postvoid residual on catheterization or ultrasound
ASSOCIATIONS	Obesity, pregnancy, vaginal delivery, prostate surgery	Bladder irritation from UTI, stones, tumors, pelvic radiation	Urinary retention, polyuria (eg, diabetes), bladder outlet obstruction (eg, BPH), spinal cord injury
TREATMENT	Pelvic floor muscle strengthening (Kegel) exercises, weight loss, pessaries	Pelvic floor physical therapy, bladder training (timed voiding, distraction or relaxation techniques), antimuscarinics (eg, oxybutynin for overactive bladder), mirabegron/vibegron	Catheterization, relieve obstruction (eg, α-blockers for BPH)

Acute cystitis

Inflammation of urinary bladder. Presents as suprapubic pain, dysuria, urinary frequency, urgency.

Systemic signs (eg, high fever, chills) are usually absent.

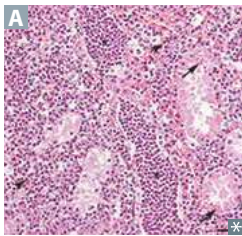
Risk factors include female sex (short urethra), sexual intercourse, indwelling catheter, diabetes mellitus, impaired bladder emptying.

Causes:

- *E coli* (most common)
- *Staphylococcus saprophyticus*—seen in sexually active young women (*E coli* is still more common in this group)
- *Klebsiella*
- *Proteus mirabilis*—urine has ammonia scent
- *Enterococcus* spp.

Labs: ⊕ leukocyte esterase. ⊕ nitrites (indicates presence of Enterobacteriaceae). Sterile pyuria (pyuria with ⊖ urine cultures) could suggest urethritis by *Neisseria gonorrhoeae* or *Chlamydia trachomatis*.

Treatment: antibiotics (eg, TMP-SMX, nitrofurantoin).

Pyelonephritis**Acute pyelonephritis**

Neutrophils infiltrate renal interstitium **A**. Affects cortex with relative sparing of glomeruli/vessels.

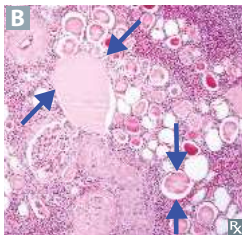
Presents with fevers, flank pain (costovertebral angle tenderness), nausea/vomiting, chills.

Causes include ascending UTI (*E coli* is most common), hematogenous spread to kidney. Presents with WBCs in urine +/- WBC casts. CT would show striated parenchymal enhancement.

Risk factors include indwelling urinary catheter, urinary tract obstruction, vesicoureteral reflux, diabetes mellitus, pregnancy (progesterone-mediated ↓ in ureter tone and compression by gravid uterus).

Complications include chronic pyelonephritis, renal papillary necrosis, perinephric abscess (with possible posterior spread to adjacent psoas muscle), urosepsis.

Treatment: antibiotics.

Chronic pyelonephritis

The result of recurrent or inadequately treated episodes of acute pyelonephritis. Typically requires predisposition to infection such as vesicoureteral reflux or chronically obstructing kidney stones.

Coarse, asymmetric corticomedullary scarring, blunted calyces. Tubules can contain eosinophilic casts resembling thyroid tissue **B** (thyroidization of kidney).

Acute kidney injury

	Prerenal azotemia	Intrinsic renal failure	Postrenal azotemia
ETIOLOGY	Hypovolemia ↓ cardiac output ↓ effective circulating volume (eg, HF, liver failure)	Tubules and interstitium: ▪ Acute tubular necrosis (ischemia, nephrotoxins) ▪ Acute interstitial nephritis Glomerulus: ▪ Acute glomerulonephritis Vascular: ▪ Vasculitis ▪ Hypertensive emergency ▪ TTP-HUS	Stones BPH Neoplasm Congenital anomalies
PATHOPHYSIOLOGY	↓ RBF → ↓ GFR → ↑ reabsorption of Na ⁺ /H ₂ O and urea	In ATN, patchy necrosis → debris obstructing tubules and fluid backflow → ↓ GFR	Outflow obstruction (bilateral)
URINE OSMOLALITY (mOsm/kg)	>500	<350	Varies
URINE Na ⁺ (mEq/L)	<20	>40	Varies
FE _{Na}	<1%	>2%	Varies
SERUM BUN/Cr	>20	<15	Varies

Acute interstitial nephritis

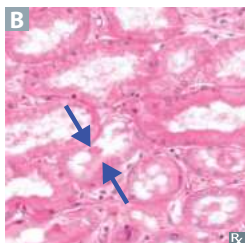
Also called tubulointerstitial nephritis. Acute interstitial renal inflammation. Pyuria (classically eosinophils) and azotemia occurring after administration of drugs that act as haptens, inducing hypersensitivity (eg, diuretics, NSAIDs, penicillin derivatives, proton pump inhibitors, rifampin, quinolones, sulfonamides). Less commonly may be 2° to other processes such as systemic infections (eg, *Mycoplasma*) or systemic inflammatory disorders (eg, Sjögren syndrome, SLE, sarcoidosis).

Associated with fever, rash, pyuria, hematuria, and costovertebral angle tenderness, but can be asymptomatic.

Remember the causes of inflammation to your

DRAINS:

- Diuretics
- Rifampin
- Antibiotics (penicillins and cephalosporins)
- Proton pump Inhibitors
- NSAIDs
- Sulfa drugs

Acute tubular necrosis

Most common cause of intrinsic acute kidney injury in hospitalized patients. Spontaneously resolves in many cases. Can be fatal, especially during initial oliguric phase. $\uparrow \text{FE}_{\text{Na}}$.

Key finding: granular casts (often muddy brown in appearance) **A**.

3 stages:

1. Inciting event
2. Maintenance phase—oliguric; lasts 1–3 weeks; risk of hyperkalemia, metabolic acidosis, uremia
3. Recovery phase—polyuric; BUN and serum creatinine fall; risk of hypokalemia and renal wasting of other electrolytes and minerals

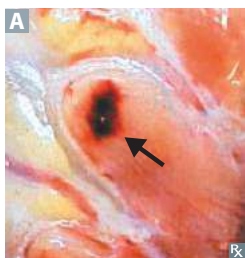
Can be caused by ischemic or nephrotoxic injury:

- Ischemic— 2° to $\downarrow$ renal blood flow (eg, prerenal azotemia). Results in death of tubular cells that may slough into tubular lumen **B** (PT and thick ascending limb are highly susceptible to injury).
- Nephrotoxic— 2° to injury resulting from toxic substances (eg, aminoglycosides, radiocontrast agents, lead, cisplatin, ethylene glycol, uric acid in tumor lysis syndrome), myoglobinuria (rhabdomyolysis), hemoglobinuria. PTs are particularly susceptible to injury.

Diffuse cortical necrosis

Acute generalized cortical infarction of both kidneys. Likely due to a combination of vasospasm and DIC.

Associated with obstetric catastrophes (eg, placental abruption), septic shock.

Renal papillary necrosis

Sloughing of necrotic renal papillae **A** $\rightarrow$ gross hematuria. May be triggered by recent infection or immune stimulus.

Associated with:

- Sickle cell disease or trait
- Acute pyelonephritis
- Analgesics (eg, NSAIDs)
- Diabetes mellitus

SAAD papa with **papillary** necrosis.

Consequences of renal failure

Decline in renal filtration can lead to excess retained nitrogenous waste products and electrolyte disturbances.

Consequences (**MAD HUNGER**):

- **M**etabolic **A**cidosis
- **D**yslipidemia (especially $\uparrow$ triglycerides)
- **H**igh potassium
- **U**remia
- **N** a^+ /**H** $_2$ **O** retention (HF, pulmonary edema, hypertension)
- **G**rowth retardation and developmental delay
- **E**rythropoietin deficiency (anemia)
- **R**enal osteodystrophy

2 forms of renal failure: acute (eg, ATN) and chronic (eg, hypertension, diabetes mellitus, congenital anomalies).

Incremental reductions in GFR define the stages of chronic kidney disease.

Normal phosphate levels are maintained during early stages of CKD due to $\uparrow$ levels of fibroblast growth factor 23 (FGF23), which promotes renal excretion of phosphate. “**FGF23** fights **f**(ph)osphate.”

Uremia—syndrome resulting from high serum urea. Can present with **P**ericarditis, **E**ncephalopathy (seen with asterixis), **A**norexia, **N**ausea (pronounce “**Ure-PEAN**” [European]).

Renal osteodystrophy

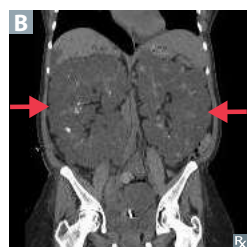
Hypocalcemia, hyperphosphatemia, and failure of vitamin D hydroxylation associated with chronic kidney disease → 2° hyperparathyroidism → 3° hyperparathyroidism (if 2° poorly managed). High serum phosphate can bind with Ca^{2+} → tissue deposits → ↓ serum Ca^{2+} .
 ↓ $1,25\text{-(OH)}_2\text{D}_3$ → ↓ intestinal Ca^{2+} absorption. Causes subperiosteal thinning of bones.

Renal cyst disorders**Autosomal dominant polycystic kidney disease**

Numerous cysts in cortex and medulla **A** causing bilateral enlarged kidneys ultimately destroy kidney parenchyma. Presents with combinations of flank pain, hematuria, hypertension, urinary infection; progressive renal failure in ~ 50% of individuals.

Mutation in genes encoding polycystin protein: PKD1 (85% of cases, chromosome 16) or PKD2 (15% of cases, chromosome 4). Complications include chronic kidney disease and hypertension (caused by ↑ renin production). Associated with berry aneurysms, mitral valve prolapse, benign hepatic cysts, diverticulosis.

Treatment: If hypertension or proteinuria develops, treat with ACE inhibitors or ARBs.

Autosomal recessive polycystic kidney disease

Mutation in *PKHD1* encoding fibrocystin. Cystic dilation of collecting ducts **B**. Often presents in infancy, and may be seen on prenatal ultrasound. Associated with congenital hepatic fibrosis. Significant oliguric renal failure in utero can lead to Potter sequence. Concerns beyond neonatal period include systemic hypertension, progressive renal insufficiency, and portal hypertension from congenital hepatic fibrosis.

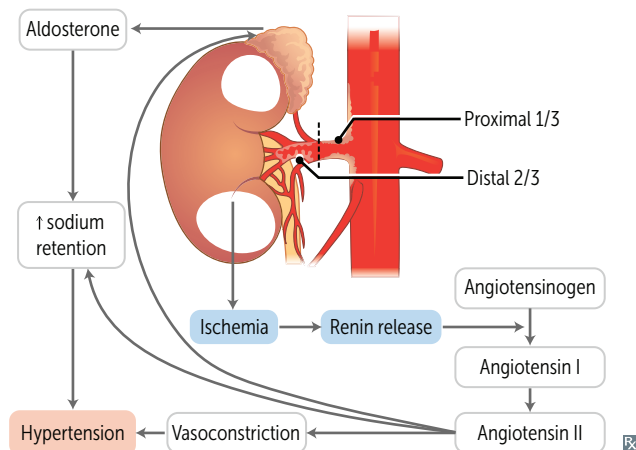
Autosomal dominant tubulointerstitial kidney disease

Also called medullary cystic kidney disease. Causes tubulointerstitial fibrosis and progressive renal insufficiency with inability to concentrate urine. Medullary cysts usually not visualized; smaller kidneys on ultrasound. Poor prognosis.

Simple vs complex renal cysts

Simple cysts are filled with ultrafiltrate (anechoic on ultrasound). Very common and account for majority of all renal masses. Found incidentally and typically asymptomatic.
 Complex cysts, including those that are septated, enhanced, or have solid components on imaging require follow-up or removal due to possibility of renal cell carcinoma.

Renovascular disease



Unilateral or bilateral renal artery stenosis (RAS) → ↓ renal perfusion → ↑ renin → ↑ angiotensin → HTN. Most common cause of 2° HTN in adults.

Main causes of RAS:

- Atherosclerotic plaques: proximal 1/3 of renal artery, usually in older males, smokers.
- Fibromuscular dysplasia: distal 2/3 of renal artery or segmental branches, usually young or middle-aged females

For unilateral RAS, affected kidney can atrophy → asymmetric kidney size. Renal venous sampling will show ↑ renin in affected kidney, ↓ renin in unaffected kidney.

For bilateral RAS, patients can have a sudden rise in creatinine after starting an ACE inhibitor, ARB, or renin inhibitor, due to their interference on RAAS-mediated renal perfusion.

Can present with severe/refractory HTN, flash pulmonary edema, epigastric/flank bruit. Patients with RAS may also have stenosis in other large vessels.

Renal cell carcinoma

Polygonal clear cells **A** filled with accumulated lipids and carbohydrate. Often golden-yellow **B** on gross pathology, due to ↑ lipid content. Originates from PCT → invades renal vein (may develop varicocele if left sided) → IVC → hematogenous spread → metastasis to lung, bone, and liver.

Manifests with flank pain, palpable mass, hematuria (classic triad) as well as anemia, 2° polycythemia (less common), fever, weight loss.

Treatment: surgery/ablation for localized disease. Immunotherapy (eg, ipilimumab) or targeted therapy for metastatic disease, rarely curative. Resistant to radiation and chemotherapy.

Most common 1° renal malignancy **C**.

Most common in males 50–70 years old, ↑ incidence with tobacco smoking and obesity.

Associated with paraneoplastic syndromes, eg, PTHrP, Ectopic EPO, ACTH, Renin (“PEAR”-aneoplastic).

Clear cell (most common subtype) associated with gene deletion on chromosome 3 (sporadic, or inherited as von Hippel-Lindau syndrome).

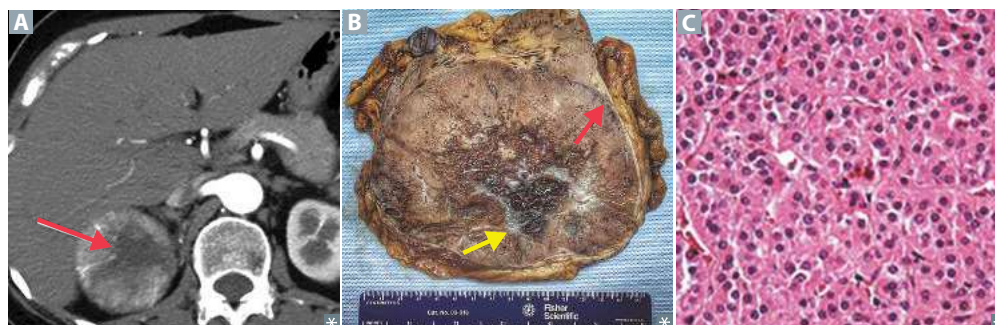
RCC = 3 letters = chromosome 3 = associated with VHL (also 3 letters).



Renal oncocytoma

Benign epithelial cell tumor **A** arising from collecting ducts (red arrow in **B** points to well circumscribed mass; yellow arrow points to central scar). Large eosinophilic cells with abundant mitochondria without perinuclear clearing **C** (vs chromophobe renal cell carcinoma). Presents with painless hematuria, flank pain, abdominal mass.

Often resected to exclude malignancy (eg, renal cell carcinoma).

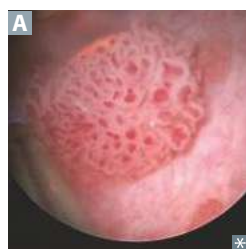
**Nephroblastoma**

Also called Wilms tumor. Most common renal malignancy of early childhood (ages 2–4). Contains embryonic glomerular structures. Most often present with large, palpable, unilateral flank mass **A** and/or hematuria and possible HTN.

Can be associated with loss-of-function mutations of tumor suppressor genes *WT1* or *WT2* on chromosome **11** (**W11**ms tumor).

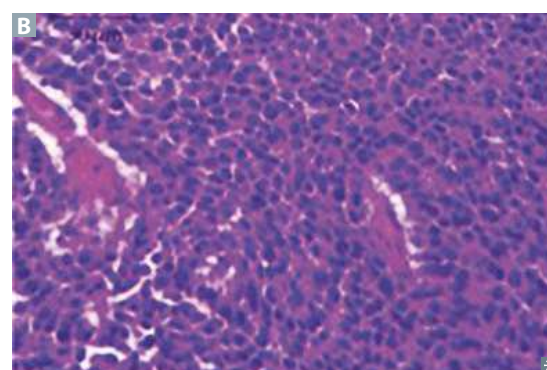
May be a part of several syndromes:

- **WAGR complex**—Wilms tumor, Aniridia (absence of iris), Genitourinary malformations, Range of developmental delays (*WT1* deletion)
- **Denys-Drash syndrome**—Wilms tumor, Diffuse mesangial sclerosis (early-onset nephrotic syndrome), Dysgenesis of gonads (male pseudohermaphroditism), *WT1* mutation
- **Beckwith-Wiedemann syndrome**—Wilms tumor, increase in organ size (organomegaly), tongue enlargement (macroglossia), hemihyperplasia (imprinting defect causing genetic overexpression, associated with *WT2* mutation), omphalocele

Urothelial carcinoma of the bladder

Also called transitional cell carcinoma. Most common tumor of urinary tract system (can occur in renal calyces, renal pelvis, ureters, and bladder) **A B**. Can be suggested by Painless hematuria (no casts).

Associated with problems in your Pee **SAC**: Tobacco Smoking, Aromatic amines (found in dyes), Cyclophosphamide.

**Squamous cell carcinoma of the bladder**

Chronic irritation of urinary bladder → squamous metaplasia → dysplasia and squamous cell carcinoma.

Risk factors include **4 S's**: *Schistosoma haematobium* infection (Middle East), chronic cystitis (“systitis”), smoking, chronic nephrolithiasis (stones). Presents with painless hematuria (no casts).